

Reader: I shall try to present the structure of the proof as a diagram (Fig. 30)

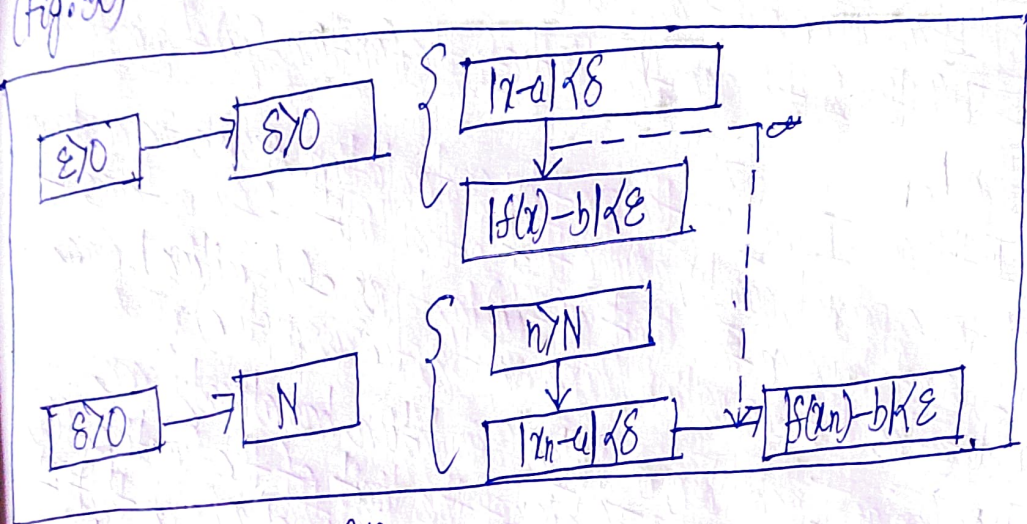


Fig. 30

Author: Your diagram is correct. Will you expand on it.

Reader: The first step of the proof: we find for an arbitrary $\varepsilon > 0$ a number $\delta > 0$ such that $|f(x)-b| < \varepsilon$ if $|x-a| < \delta$.
 The second step of the proof: we take δ selected at the first step, choose a sequence (x_n) convergent to a , and find a number N such that $|x_n-a| < \delta$ for all $n > N$. Having in mind the arguments used at the first step, we conclude that $|f(x_n)-b| < \varepsilon$ if $|x_n-a| < \delta$.
 We have thus found for an arbitrary $\varepsilon > 0$ a number N such that $|f(x_n)-b| < \varepsilon$ for all $n > N$. This completes the proof.

Author: Correct. In conclusion I want to emphasize several essential points on which the proof hinges. We know that $|f(x)-b| < \varepsilon$ for any x from the δ -neighbourhood of a . Since a sequence (x_n) is convergent to a , all x_n (the whole infinite "tail" of the sequence (x_n) starting from a certain number $N+1$) are contained inside the δ -neighbourhood of point a . It then follows that all $f(x_n)$ (the whole infinite "tail" of the sequence $[f(x_n)]$ starting from the same number $N+1$) are contained inside the interval $[b-\varepsilon, b+\varepsilon]$. This proves that the sequence $[f(x_n)]$ converges to b .

Reader: I understand.